

AI for social impact (PS 10)

S.A.M.A.R.T.H

Social Action management & Real Time Help

PRESENTED BY PENGUINS 

MEMBERS:-

Astha

Dhruv

Sanyogita

Prachi

What is the problem?

- Crores of Indians miss the government job opportunities.
- Information is scattered
- Document confusion

Why does this problem matter?

- Missed opportunities
- Wasted effort or time
- Unequal access

Who is affected?

- Students
- Job seekers
- Communities and organisations

KEY LIMITATIONS

- Information overload
- Poor personalization
- Difficult navigation
- Scattered information
- Time consuming Manual filtering

EXISTING SOLUTIONS

- SarkariResult.com
- Google & YouTube
- Official Recruitment portals

REFERENCES

- Government department websites
- Rozgar samachar
- Official government recruitment portals

OUR SOLUTION

**Upload once,
Not fill forever**

**Personalized
Opportunities**

Timely Updates

S.A.M.A.R.T.H

**AI Powered
Requirement
Extraction**

**Save Time and
Effort**

**Simple &
Accessible**

SYSTEM ARCHITECTURE

Upload

Resume, Degree, ID proof, Experience letters

Extract

PDF / OCR → AI reads documents → structured profile

Match

Fetch live government jobs → compare profile vs. requirements

Deliver

✓ Eligibility · 📄 Documents needed · 🔗 Apply link · 📌 Source

TECHNOLOGY & IMPLEMENTATION

Backend

- Python 3.12 — core language for the entire backend
- FastAPI — REST API framework; auto-generates OpenAPI docs
- Uvicorn — ASGI server with hot-reload for development
- Pydantic — request/response validation

Frontend

- Vanilla HTML5 + CSS3 — no framework, no build step, single file
- Vanilla JavaScript (ES6+) — fetch API, FormData, drag-and-drop events
- Font Awesome — icon set
- Google Fonts (Inter) — typography

Document processing

- Tesseract OCR 5.3 — extracts text from Scanned PDFs and images
- pypdf / PyPDF2 — extracts text from native PDF files
- Pillow (PIL) — image loading and format handling
- Multipart file upload — browser → backend via FormData

Data Sources & Networking

- Requests — HTTP client for fetching notification pages
- BeautifulSoup4 — HTML parsing and text extraction
- Cloudflare-resistant headers — browser-like User-Agent to improve scrape success rate

WORKING PROTOTYPE

2 REVIEW YOUR EXTRACTED PROFILE

S.A.M.A.R.T.H. read your documents. Check and correct anything below before matching.

Full name

Dhruv Vishwakarma

Email

dhruvjeron1S@gmail.com

Phone

Date of birth (YYYY-MM-DD)

14/12/2006

Highest qualification

Bachelors

Field of study

Computer Science Engineering

Years of experience

0

Location

Prithvipur, Niwari, Madhya Pradesh

Category

OBC

Age

19

Skills

C, C++, Python, PyQt5, NumPy, JSON, Pygame, Git, GitHub, Pyinstaller, Linux, Windows, Data Structures & Algorithms, OOP, Problem Solving, Debugging, Packag

Documents detected in your upload

Resume, Caste certificate, Domicile certificate, Income certificate

AI summary

Computer Science undergraduate with a strong interest in system-level understanding, problem solving, and building real-world applications. Passionate about Linux, open-source technologies, and exploring the intersection of software, hardware, and science.

 Extract profile from documents

 Find matching opportunities

 Reset

 Profile extracted from 4 file(s) Review below, then click Find matching opportunities.

RESULTS & TESTING

🔍 Opportunities for Dhruv Vishwakarma

Showing 5 · 0 eligible

Sarkari Result

🏢 Government of India

⚠️ Needs verification

🎓 —

👤 Age ≤ —

DOCUMENTS YOU ALREADY HAVE

None matched from your upload

STILL NEEDED

All required documents present ✓

📅 See notification

Rs See notification

View notification →

📄 Source: SarkariResult

UPSSSC Senior Instructor Online Form 2026 | Last Date : 05/10/2026

🏢 Government of India

⚠️ Needs verification

🎓 —

👤 Age ≤ —

DOCUMENTS YOU ALREADY HAVE

None matched from your upload

STILL NEEDED

All required documents present ✓

📅 See notification

Rs See notification

View notification →

📄 Source: SarkariResult

ISRO ICRB Assistant and Other Post Admit Card 2026

🏢 Government of India

⚠️ Needs verification

🎓 —

👤 Age ≤ —

DOCUMENTS YOU ALREADY HAVE

None matched from your upload

STILL NEEDED

All required documents present ✓

📅 See notification

Rs See notification

View notification →

📄 Source: SarkariResult

SSC CPO SI CAPF Online Form 2026 | Last Date : 30/09/2026

🏢 Government of India

⚠️ Needs verification

🎓 —

👤 Age ≤ —

DOCUMENTS YOU ALREADY HAVE

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STILL NEEDED

All required documents present ✓

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View notification →

📄 Source: SarkariResult

Bank of India BOI Specialist Officers SO Online Form 2026 | Last Date : 25/09/2026

🏢 Government of India

⚠️ Needs verification

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👤 Age ≤ —

DOCUMENTS YOU ALREADY HAVE

None matched from your upload

STILL NEEDED

All required documents present ✓

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View notification →

📄 Source: SarkariResult

FUTURE SCOPE

- **Add more job categories and government schemes**
- **Include regional language support for wider reach**
- **Expand to include internship**

SCALIBILITY

- **Can be expanded to serve users across all States**
- **Can integrate with multiple government and private platforms**

CONCLUSION

- **S.A.M.A.R.T.H. addresses the real problem of information overload and lack of personalization in job searches.**
- **It makes opportunity hunting simple, structured, and user-friendly.**
- **By combining trusted sources, AI-powered matching, and a personalized profile, it saves time and reduces opportunities.**
- **This can create a bigger impact by making government opportunities more accessible, especially for students and first-time job seekers.**